

Project Title	Human Voice Classification and Clustering
Skills take away From This Project	Python scripting, Machine Learning, Pre Processing, Feature Engineering, Feature Selection, Clustering, Classification, Model Evaluation, Streamlit
Domain	Speech Processing, Machine Learning

Problem Statement

Develop a machine learning system that classifies and clusters human voice profiles using pre-extracted numerical audio features. The workflow includes preprocessing the dataset, performing clustering to identify natural groupings, training classification models to predict predefined voice categories, and evaluating model performance. The final application will provide a Streamlit interface where users can manually enter numerical feature values and receive both cluster assignment and classification predictions in real time.

Business Use Cases

- 1. Speaker Identification:**
 - Identify individuals based on their voice features.
- 2. Gender Classification:**
 - Classify voices as male or female for various applications like call center analytics.
- 3. Speech Analytics:**
 - Extract insights from audio data for industries such as media, security, and customer service.
- 4. Assistive Technologies:**
 - Improve accessibility solutions by analyzing voice patterns.

Approach

Data Preparation:

- Use the provided dataset containing extracted voice features.
- Split the dataset into training, validation, and test sets.

Data Cleaning and Preprocessing:

- Handle missing or inconsistent data.
- Normalize numerical features for consistency.

Exploratory Data Analysis (EDA):

- Visualize feature distributions and correlations.
- Analyze patterns in spectral and pitch-related features.

Model Development:

- Apply clustering techniques (K-Means, DBSCAN) to group similar voice samples.
- Train classification models (Random Forest, SVM, Neural Networks) to predict gender labels.
- Experiment with feature selection and hyperparameter tuning.

Model Evaluation and Comparison:

- Use metrics like accuracy, precision, recall, F1-score, silhouette score (for clustering), and confusion matrix.
- Compare model performance and select the best-performing approach.

Application Development:

- Build a Streamlit-based interface for users to upload audio samples and receive predictions.

Data Flow and Architecture

Data Collection & Preprocessing:

- Use the Extracted data and normalize spectral, pitch, and MFCC features from audio files.

Processing Pipeline:

- Implement feature engineering, dimensionality reduction, and normalization.

Model Training:

- Train machine learning models using Scikit-learn or TensorFlow.
- Save trained models for deployment.

Deployment:

- Create a Streamlit-based front-end for users to interact with the model.

A Streamlit application will be developed with clear navigations, including:

1. **Introduction Page** – An overview of the project, dataset, objectives, and model pipeline.
2. **EDA & Visualization Page** – Graphical exploration of numerical audio features (distributions, correlations, cluster visualizations, etc.).
3. **Classification and Clustering Page(s)** –
 - You may keep **classification and clustering on a single page** or create **two separate pages**, based on preferred UI design.
 - Users will enter **numerical feature values manually** in Streamlit to get predictions.
 - Instead of requiring all 40+ audio features, the app will use a selected set of **15–20 top features** for demonstration and model prediction.

The final Streamlit interface will allow users to input feature values, generate cluster labels, view classification predictions, and interpret model outputs interactively.

Dataset

[human voice clustering.zip](#)

Dataset Explanation:

The dataset consists of extracted audio features from human voice recordings, categorized as follows:

- **mean_spectral_centroid:** The average spectral centroid, representing the "center of mass" of the spectrum, indicating brightness.
- **std_spectral_centroid:** The standard deviation of the spectral centroid, measuring variability in brightness.
- **mean_spectral_bandwidth:** The average width of the spectrum, reflecting how spread out the frequencies are.
- **std_spectral_bandwidth:** The standard deviation of spectral bandwidth, indicating variability in frequency spread.
- **mean_spectral_contrast:** The average difference between peaks and valleys in the spectrum, indicating tonal contrast.
- **mean_spectral_flatness:** The average flatness of the spectrum, measuring the noisiness of the signal.
- **mean_spectral_rolloff:** The average frequency below which a specified percentage of the spectral energy resides, indicating sharpness.
- **zero_crossing_rate:** The rate at which the signal crosses the zero amplitude axis, representing noisiness or percussiveness.
- **rms_energy:** The root mean square energy of the signal, reflecting its loudness.
- **mean_pitch:** The average pitch frequency of the audio.
- **min_pitch:** The minimum pitch frequency.
- **max_pitch:** The maximum pitch frequency.
- **std_pitch:** The standard deviation of pitch frequency, measuring variability in pitch.
- **spectral_skew:** The skewness of the spectral distribution, indicating asymmetry.
- **spectral_kurtosis:** The kurtosis of the spectral distribution, indicating the peakiness of the spectrum.
- **energy_entropy:** The entropy of the signal energy, representing its randomness.
- **log_energy:** The logarithmic energy of the signal, a compressed representation of energy.
- **mfcc_1_mean to mfcc_13_mean:** The mean of the first 13 Mel Frequency Cepstral Coefficients (MFCCs), representing the timbral characteristics of the audio.
- **mfcc_1_std to mfcc_13_std:** The standard deviation of the first 13 MFCCs, indicating variability in timbral features.
- **label:** The target variable indicating the gender (male = 1, female = 0).

Results

By the end of this project, learners will achieve:

- A preprocessed and structured dataset ready for machine learning.
- Multiple clustering and classification models trained and evaluated.
- A deployed voice classification system accessible via a Streamlit interface.

Project Evaluation Metrics

Data Preprocessing Quality:

- How well the data is cleaned, structured, and normalized.

Model Performance:

- Accuracy, precision, recall, F1-score for classification.
- Silhouette score and cluster purity for clustering.

Application Functionality:

- User-friendly Streamlit interface for predictions.

Technical Tags

Python, Machine Learning, Pre Processing, Clustering, Classification, Streamlit, Scikit-learn, Feature Engineering, Feature Selection.

Deliverables

Data Preparation:

- Evaluate how well the dataset is cleaned and preprocessed.

Model Training:

- Assess performance across different clustering and classification models.

Evaluation Metrics:

- Compare models using classification and clustering metrics.

Application Deployment:

- Ensure the Streamlit interface is accessible and responsive.

Timeline

7-10 days

References:

Project Live Evaluation Metrics	Project Live Evaluation
EDA Guide	Exploratory Data Analysis (EDA) Guide
Capstone Explanation Guideline	Capstone Explanation Guideline
GitHub Reference	How to Use GitHub.pptx
Streamlit recordings (English)	Special session for STREAMLIT(11/...
Streamlit recordings (Tamil)	Streamlit - Tamil str_app.py
Streamlit documentation	Install Streamlit
Project Orientation (English)	Project Orientation Session : Human ...
Project Orientation (Hindi)	Project Orientation Session : Human ...

PROJECT DOUBT CLARIFICATION SESSION (PROJECT AND CLASS DOUBTS)

About Session: The Project Doubt Clarification Session is a helpful resource for resolving questions and concerns about projects and class topics. It provides support in understanding project requirements, addressing code issues, and clarifying class concepts. The session aims to enhance comprehension and provide guidance to overcome challenges effectively.

Note: Book the slot at least before 12:00 Pm on the same day

Timing: Monday-Saturday (3:30PM to 4:30PM)

Session link :

 **Project Doubt Session [DS/AIML]**

LIVE EVALUATION SESSION (CAPSTONE AND FINAL PROJECT)

About Session: The Live Evaluation Session for Capstone and Final Projects allows participants to showcase their projects and receive real-time feedback for improvement. It assesses project quality and provides an opportunity for discussion and evaluation.

Note: This form will Open only on Saturday (after 2 PM) and Sunday on Every Week

Timing:

For DS and AIML

Monday-Saturday (11:30AM to 12:30PM) and (05:30PM to 07:00PM)

Booking link : <https://forms.gle/1m2Gsro41fLtZurRA>

Evaluation Metrics : [Project Live Evaluation](#)

Project Created By	Verified By	Approved By
Vinodhini	Nehlath Harmain	Nehlath Harmain